

Optimal and Stable Framework for Robust Field-Oriented Control of PMSM using Hybrid ALO-MRSO

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ABSTRACT

This research presents a control method that combines ALO and MRSO to address the dynamic stability issue and optimal performance in permanent magnet synchronous motor (PMSM) drives. The combination of ALO-MRSO facilitates global optimization of controller parameters by efficiently exploring high-dimensional search spaces to provide asymptotic stability by confining system trajectories within specified boundaries. This hybrid optimization paradigm provides an advantage over conventional methods by not only ensuring the convergence of the optimization process to feasible solutions but also by enforcing closed-loop stability conditions. The framework is integrated within a field-oriented control (FOC) structure, in which the ALO-MRSO algorithm systematically optimizes each proportional-integral (PI) regulator within the cascade control loop. The results verified the improvements of the combined ALO-MRSO method in both transient and steady-state performance over the ALO or MRSO method based on MATLAB Simulink. The results of the optimized system achieve a rise time of 0.68 seconds, a minimal overshoot of 0.03%, and a settling time of 0.63 seconds. This paper establishes a systematic and theoretical basis for managing the exploration and exploitation balance in advanced electromechanical control of PMSM systems, enabling rapid dynamic response with exceptional stability. Therefore, a scalable approach for a robust controller design and optimal performance is achieved for a nonlinear dynamic system.

Keywords: Ant lion optimization (ALO), field-oriented control (FOC), modified rat swarm optimization (MRFO), Permanent magnet synchronous motor (PMSM).

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1. INTRODUCTION

Permanent magnet synchronous motors have gained very high popularity for a very broad range of applications due to their high efficiency, very accurate dynamic response, and high power density over the last few years [1, 2]. On the other hand, PMSMs control shows certain inherent problems given the uncertain parameters, the sensitivity to external disturbances, and variations of the system parameters during operation [3–5].

The different algorithms for PMSM drives' precise control, which are divided as classical and non-classical methods, have been addressed by the researchers. Conventionally, PI control, FOC, and DTC are largely preferred because of their simplicity and ease of implementation. They present an acceptable performance in well-tuned

operating conditions [1, 6, 7]. As for the non-classical approaches, they usually show weak robustness to parameter sensitivity and are not robust to nonlinearity, external disturbances, or parameter variations. Hence, they may not guarantee optimum performance under different operating conditions [8].

Conversely, Non-classical control methods, which include model predictive control (MPC), Sliding Mode Control (SMC), fuzzy control, neural network control, and heuristic optimization techniques have emerged to overcome these limitations [4], [9]–[15]. However, most non-classical methods suffer from complexity, uncertainties, chattering effects, large amounts of training data, and intensive computation. However, Heuristic optimization techniques, such as the Ant Lion Optimization algorithm



(ALO) and Modified Rat Swarm Optimization algorithm (MRSO), are highly efficient in optimizing the controller parameters for complex nonlinear systems, offering global search capabilities and flexibility [16]–[18]. Therefore, selecting an appropriate control strategy for PMSM drives requires a balance between complexity, robustness, computational intensity, and desirable performance outcomes.

Among the heuristic approaches, the Ant Lion Optimizer (ALO) and Modified Rat Swarm Optimization (MRSO) algorithms are proposed in this study to overcome the limitations of classical optimization techniques and address the challenges posed by non-classical optimization techniques [18], [19]. These are up-and-coming methods for resolving complex engineering problems and making them more effective tools for adjusting control parameters and improving system accuracy and robustness significantly.

The rest of this paper is organized as follows: Section II introduces the ALO and MRSO algorithms, along with their application to PMSM control. Section III explains the PMSM model and conventional control problems. Section IV provides the simulation and experimental results, and Section V concludes the paper with the primary findings and future research directions.

2. PROPOSED ALO-MRSO HEURISTIC TECHNIQUES IN THE PMSM CONTROL APPROACH

Ant Lion Optimizer is inspired by the natural hunting activities of antlions. It was based on work that mimics the interaction of antlions with ants. Because the hunting search behaviour scrupulously searches through the search space in order to avoid local optima, an antlion has the ability to be very effective in global optimisation. However, its exploitation is restricted to enhancing accurate solutions. Due to poor convergence and battling to fine tune the solution, subsequent refinement-when a promising region has been obtained is difficult. Therefore, they can become stuck if the elite does not improve [16, 17, 19].

While on the other hand, modified rat swarm optimization algorithm depends upon the social and behavioral characters of the rats. As the work presented in [18], the main factor of Exploration and Exploitation was improved by keeping a good harmony between the factors. However, the violent nature of the rats might lead to the convergence of MRSO much faster.

Therefore, using the ALO algorithm as the leading global explorer and embedding MRSO's mechanism as a powerful local exploiter to utilize the hybrid ALO-MRSO algorithm to avoid getting stuck in local optima and to find more accurate solutions, which improve local optimization and enable high-precision refinement near optimal points. The strengths of the new ALO-MRSO integrated algorithm are leveraged. Thus, ALO conducts common exploration to identify good regions in the solution space, and MRSO refines these solutions with localized precision. Therefore, the hybrid algorithm achieves a faster convergence speed and greater accuracy than single algorithms, which makes it particularly well-suited for dynamic PMSM control systems. The hybrid ALO-MRSO strength

yields a properly balanced optimization structure to achieve a robust optimization platform for PMSM control systems, ensuring an optimal control performance across different operating regimes.

The ALO is also very efficient in the realm of global optimization, mainly because it ensures a proper balance between the exploration and exploitation in the search space [16]. By adaptive radial improvements, the MRSO increases the precision of the local search, and is thus very suited to dynamic systems [18, 20]. Therefore, with the integration of the two optimization algorithms, ALO-MRSO, PI gains tuning becomes more accurate. Thus, it is guaranteed that, under an ALO-MRSO control strategy, the current ripple in PMSMs will be reduced with transient response enhanced by stability in the system.

The hybrid methodology of ALO-MRSO enhances the control performance by ensuring robustness, improving computational efficiency, and automatic self-tuning of PI gains that ultimately result in torque ripple reduction and current tracking accuracy improvement. According to the hybrid approach, there is still stability of performance under various working conditions, proving adaptability to parameter drifts and changes of loads. Finally, concerning the traditional methods of optimization, the hybrid one reduces the computational burden, considering the high convergence rate of ALO and the fact that MRSO provides very accurate results.

The ALO-MRSO hybrid method aims to maximize the control performance, enhance the robustness, and improve the computational efficiency. This is achieved through the automatic self-tuning of the PI gains, which then reduces the torque ripple and improves the current tracking accuracy. The hybrid approach ensured a stable performance under different working conditions and demonstrated the adaptability to parameter drifts and load changes. The hybrid approach reduces the computational burden over conventional optimization methods by leveraging the high convergence rate of the ALO and the accuracy of the MRSO.

The algorithm operates in three synergistic phases: Global Exploration Phase, the Transition Phase, and the Local Refinement Phase. A flow chart of these three phases is presented in Fig. 1. First, ALO is Dominant within the Global Exploration Phase. Adaptive Balancing was dominant during the transition phase. Finally, MRSO is Dominant within the local refinement phase.

ALO-MRSO hybridization has certain specific advantages from the point of view of application in PMSM control.

- Simultaneous global parameter space search and local controller adaptation
- Dynamic Balancing Mechanism: Dynamically adapts to time-varying operating conditions.
- Diversity of solution maintenance to avoid early convergence in multi-modal solution landscapes
- The goal is to effectively split the computational resources between exploration and exploitation.

This convergence can be tuned, and hence one can use a hybrid stop criterion that best blends solution stability with optimization progress. Further, this must result in a

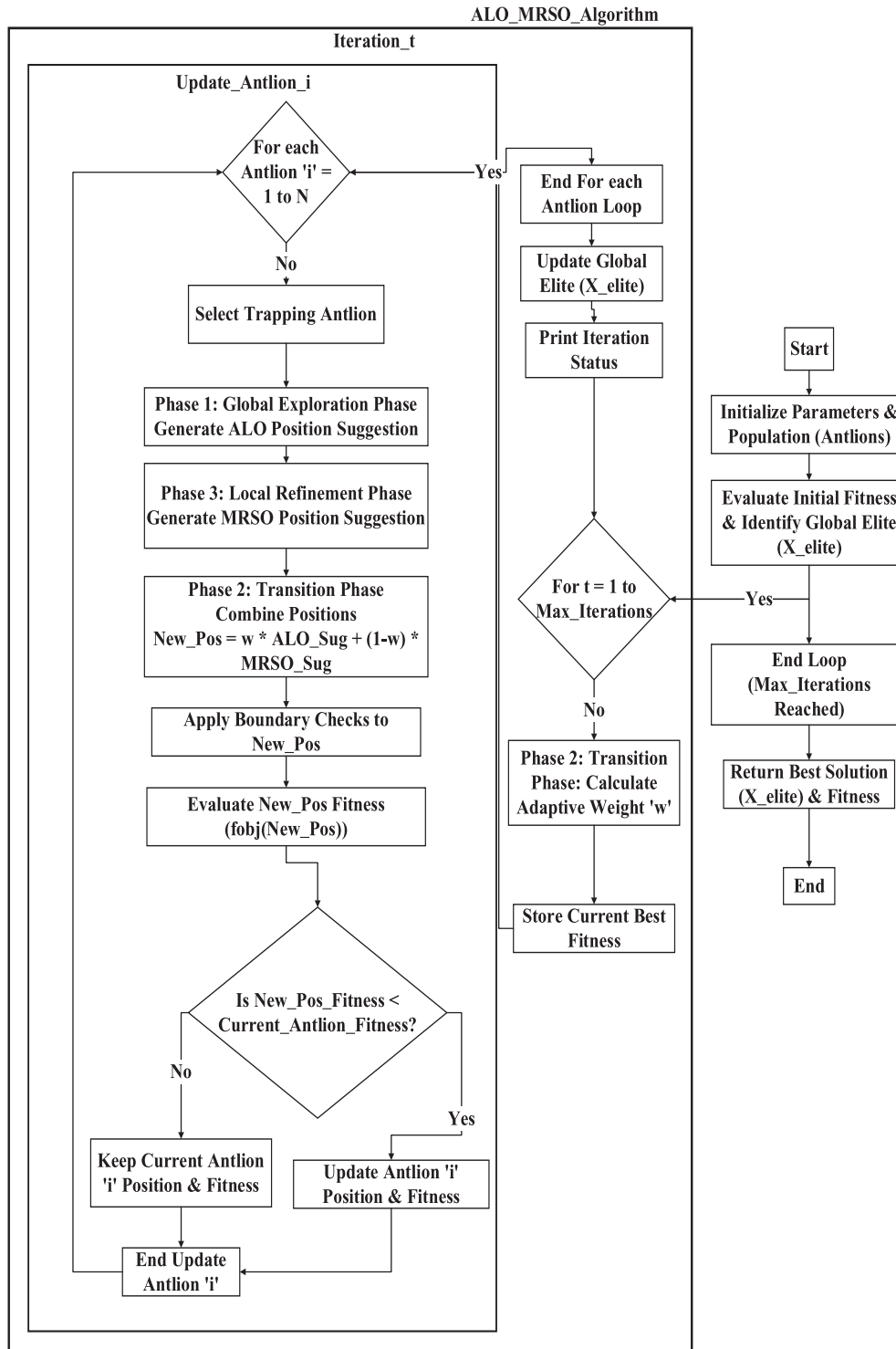


Fig. 1. Flowchart of the ALO-MRSO optimization phases process.

balanced performance across multiple PMSM operating regimes. In particular, an example of such hybrid setup is especially befitting for these difficult motor control optimization problems, especially in the case of simultaneous tuning of various controller parameters under dynamic load conditions. It is shown by the simulations that the proposed approach is valid and even outperforms the traditional PI-based and fixed-hysteresis control approach.

The validity of the proposed method is confirmed by simulations, which demonstrate its superiority over the traditional PI-based and fixed-hysteresis control approaches.

3. MATHEMATICAL REPRESENTATION OF PERMANENT MAGNET SYNCHRONOUS MOTOR (PMSM)

Its PMSM has been characterized based on stator circuit and can be effectively modelled in the d-q reference frame. On the other hand, the rotor reference frame selection is dependent mainly on modelling because it has a direct impact on generating electromagnetic force, stator voltage, and torque produced by the rotor [2, 21].

The process is simplified by reducing or eliminating some of the complex factors such as magnetic saturation, rotor saliency, and system nonlinearities, which are

ignored for better control design [21, 22]. The magnetic field distribution and back electromotive force are purely sinusoidal. Besides, harmonic components in the air gap along with iron losses are also neglected. These simplifications make the model of PMSM more manageable and allow faster computations and analysis, helping towards efficient control design [14, 23].

The dynamic behavior of a PMSM can be mathematically modelled in the rotor reference frame (d-q axes) using the following fundamental equation [24]–[26].

1. Voltage and Electromagnetic Torque Equations is shown in (1):

$$\begin{aligned} v_q &= Ri_q + L_q \frac{d}{dt} i_q + \omega_r (L_d i_d + \psi_m) \\ v_d &= Ri_d + L_d \frac{d}{dt} i_d - \omega_r (L_q i_q) \\ T_e &= \frac{3}{2} \times \frac{P}{2} (\psi_m i_q + (L_d - L_q) i_d i_q) \end{aligned} \quad (1)$$

In a surface-mounted PMSM, $L_d = L_q = L$, and the torque equation can be expressed as in (2):

$$T_e = \frac{3}{2} \times \frac{P}{2} (\psi_m i_q) \quad (2)$$

Assuming that magnetic flux linkage is constant, the torque is linearly dependent on the current along the q-axis. The Mechanical Motion equation for the electromagnetic torque can be expressed as in (3):

$$J \frac{d\omega_r}{dt} = T_e - T_L - B\omega_r \quad (3)$$

The Rotor Position Relationship can be represented as (4):

$$\theta_r = \int \omega_r dt \quad (4)$$

4. FIELD-ORIENTED CONTROL (FOC) FOR PMSM DRIVES

Field-Oriented Control or vector control is an advanced control scheme that decouples the torque and flux components of a PMSM. It performs such a decoupling by transforming the motor variables into a rotating reference frame, the so-called d-q axes, referenced with respect to the rotor flux [6, 7, 27]. In this framework, the id-axis current controls the magnetic flux of the motor while the iq-axis current independently controls torque generation, hence such concepts allow independent and precise control over both vital parameters. By this transformation, using Clarke and Park transformations, nonlinear dynamics of the motor are effectively linearized, which enables FOC to achieve high-performance behavior with low-ripple torque response and high efficiency in a large speed range. The block diagram of the field-oriented control is presented in Fig. 2.

Accurate rotor position feedback is indispensable to ensure proper field orientation. FOC can provide MTPA operation and allow flux weakening in high-speed applications. Hence, in applications like electric vehicles, robotics, or industrial automation, the use of FOC is preferred in many systems for dynamic performance [6, 14, 27].

5. OPTIMIZATION OF A MULTI-OBJECTIVE COST FUNCTION

In PMSM control, the cost function of optimization is designed as a mathematical criterion for evaluating and minimizing the control errors so as to satisfy the system constraints. In heuristic optimization algorithms, the cost function usually encapsulates multiple performance objectives. Each component of this function focuses on a different aspect of performance combined in one metric, J . They are described below [28–31]:

- **Tracking Error Minimization:** This equation quantifies the primary control objective, ensuring that the actual output of the motor aligns with the desired reference value.
- **Torque Ripple Reduction:** This equation aims to minimize the torque ripple, thereby ensuring smooth and stable motor operation as shown in (5).

$$J_2 = k_4 \sum (T_e - T_e^*)^2 \quad (5)$$

- **Energy Efficiency:** This equation directly targets the energy efficiency of the motor as shown in (6).

$$J_3 = k_5 \int (i_d^2 + i_q^2) dt \quad (6)$$

- **Constraint Handling (Soft Constraints):** This equation acts as a penalty function to handle operational constraints, particularly the voltage limits of the inverter as shown in (7).

$$\begin{aligned} J_4 &= k_6 \sum \max(0, |v_d| - v_{max})^2 \\ &+ k_7 \sum \max(0, |v_q| - v_{max})^2 \end{aligned} \quad (7)$$

- **Composite Cost Function:** This final equation combines all the individual objectives into a single scalar value as shown in (8).

$$J_t = \sum_{i=1}^4 j_i \quad (8)$$

Weight Adaptation: Heuristic algorithms update the weights dynamically in a heuristic way, reflecting the relative importance of objectives under different conditions, such as high torque or at high speed. Multi-objective optimization seeks a balance by considering, for instance, competing goals of tracking and efficiency via Pareto-front analysis.

In detail, ALO-MRSO and similar algorithms tune controller gains in real time by iteratively minimizing J with examples of parameters as PI parameters and hysteresis bands. The proposed framework will allow for robust PMSM control, optimized transient response, high steady-state accuracy, energy efficiency, and satisfaction of hardware constraints.

6. SIMULATION RESULTS

This paper proposes an FOC strategy, which is implemented in a MATLAB/Simulink environment. It consists

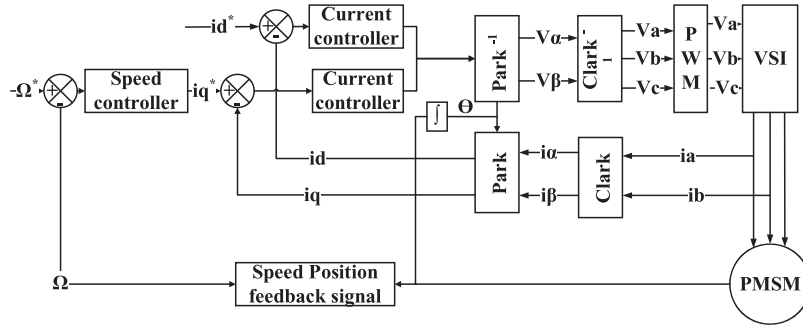


Fig. 2. Block diagram of the FOC.

TABLE I: THE OPTIMAL PROPORTIONAL AND INTEGRAL GAINS FOR DIFFERENT METHODS

Method	Controller	Kp	Ki
ALO	Speed	7.2	17.3
	q-axis current	2.5	11.8
	d-axis current	1.4	8.6
MRSO	Speed	9.4	30.5
	q-axis current	3.6	13.7
	d-axis current	2.7	33.4
ALO-MRSO	Speed	1.5	55.8
	q-axis current	0.9	13.5
	d-axis current	0.9	11.2

of an outer speed control loop and inner current regulation loops. The outer speed control loop produces a torque-producing current reference signal. A reference is achieved by measuring the error deviation between the commanded speed and the actual motor speed via a PI controller. Similarly, the inner control loops are the decoupled d-axis and q-axis current controllers. In each axis, a different PI controller regulates the current. The d-axis is usually set to zero in surface-mounted PMSMs to allow the torque to be generated most efficiently. In this regard, the primary goal of the proposed model is to optimize the PI controller's gain by using three different metaheuristic optimization approaches: Ant Lion Optimizer, Modified Rat Swarm Optimization, and the novel hybrid ALO-MRSO approach. For enabling precise interaction between the controller and the three-phase motor system, essential reference frame transformations

(abc-dq and dq-abc conversions) were considered. Further, the two-level, three-phase voltage source inverter-VSI is driven by 500V DC link which converts the control voltage signals into pulse-width modulated waveforms. This block models the electromagnetic and mechanical dynamics of the motor and calculates variables such as stator currents, electromagnetic torque, and rotor speed. In this paper, the objective function of the system performance was quantitatively evaluated as the weighted fitness comprising 40% error metrics, 30% overshoot, 20% rise time, and 10% ripple factor. For dynamic behavior assessment of the system, torque ripple measurements can be considered. As for primary contributions, the integration of metaheuristic optimization via MATLAB subroutine permits systematic tuning of control parameters yielding significant improvements in speed tracking, torque ripple reduction, and overall dynamic response. This simulation gives a very robust framework to investigate the effectiveness of advanced control strategies combined with optimization techniques in PMSM drive systems. The proportional and integral gains of the speed loop and the inner d-axis and q-axis current control loops are provided in Table I.

The performances of different metaheuristic methods, namely ALO, MRSO, and hybrid ALO-MRSO, are represented in the results. Figures present the performances measured at different step changes in the speed command and provide details about its ability to track various reference speeds precisely and other parameters like load torque, motor torque, and motor current. Performance

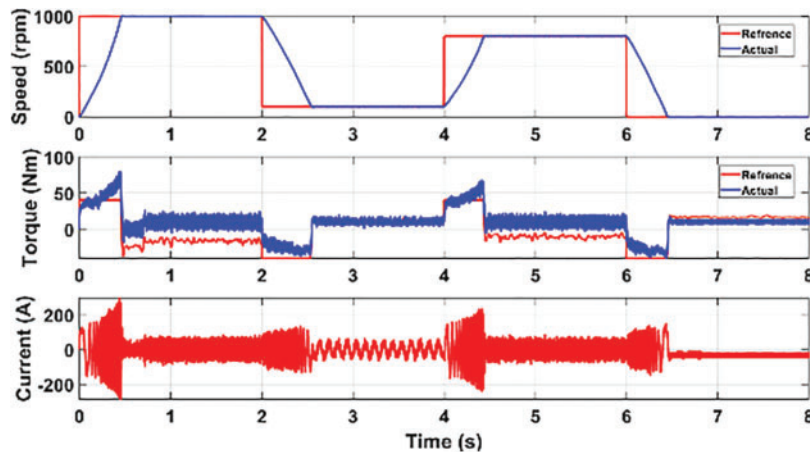


Fig. 3. Step response performance of the ALO method.

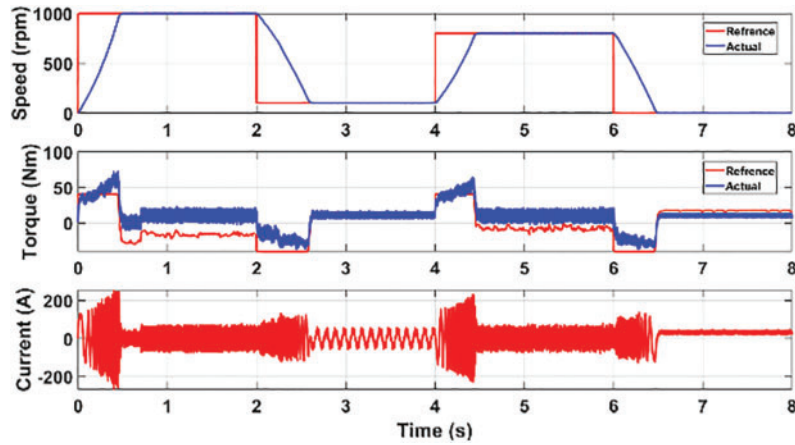


Fig. 4. Step response performance of the MRSO method.

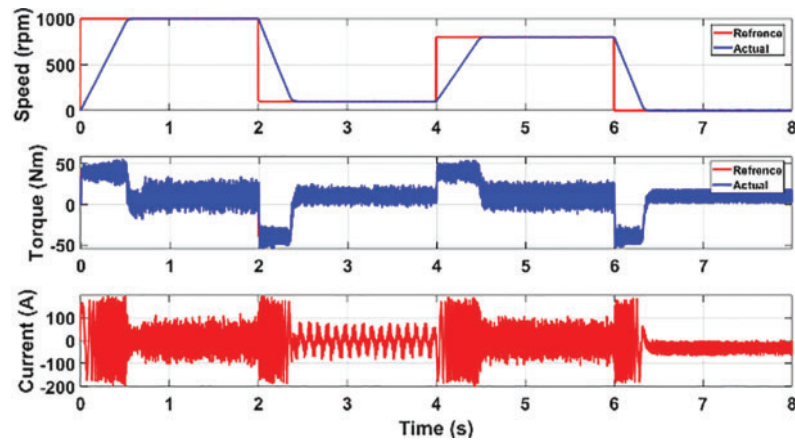


Fig. 5. Step response performance of the ALO-MRSO method.

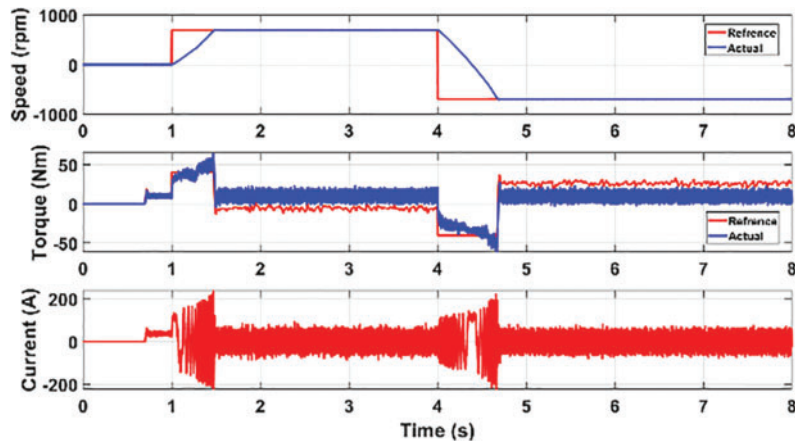


Fig. 6. Four-quadrant operation of the ALO method.

measured with the optimization method ALO at various step changes is presented in Fig. 3, while those from MRSO and hybrid ALO-MRSO are represented in Fig. 4 and Fig. 5, respectively. Corresponding to the ALO-optimised controller, Fig. 3 gives a relatively fast speed tracking response of about 0.75 s rise time and an overshoot of about 0.08 % during acceleration. The system quickly reaches the steady state; however, there exist small oscillations of torque and current during transient phases.

Fig. 4 shows the response of the tuning based on MRSO. It is evident that there are notable improvements: this method has improved torque smoothness during the load

transitions with a slightly faster rise time of about 0.7 seconds and reduced overshoot of about 0.05%. Additionally, the current response has also attenuated high-frequency noise more effectively, which indicates increased robustness and superior regulation in the current loop. The MRSO method appears to be more effective in reducing ripples in torque and dampening lower spikes in current, reflecting improvements for global search ability in optimizing PI gains.

Fig. 5 reveals the outstanding performance of the hybrid ALO-MRSO approach, which provides the best performance in contrast to the two other optimization methods.

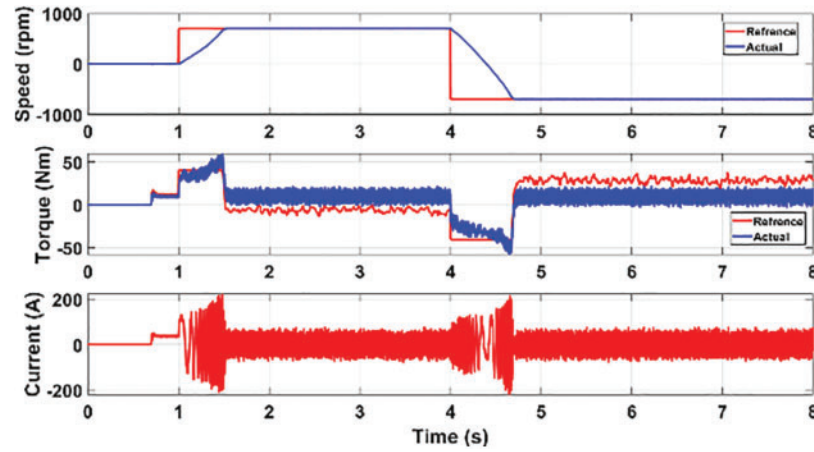


Fig. 7. Four-quadrant operation of the MRSO method.

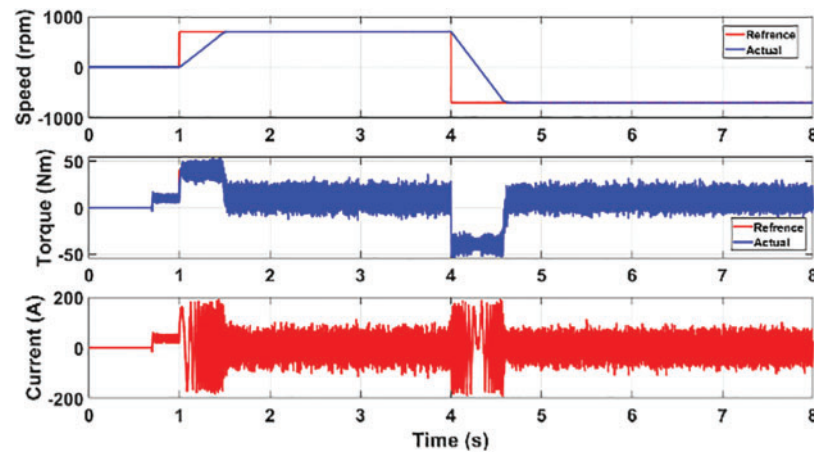


Fig. 8. Four-quadrant operation of the ALO-MRSO method.

The speed curve has an almost negligible steady-state error and a very small overshoot that is less than 0.03%, with the fastest rise time of all methods at about 0.68 s. Moreover, the torque waveform was very smooth and the current response showed the smallest ripple and noise over all phases of operation. This further confirms that the combination of the exploration ability of the ALO with the exploitation strength of MRSO indeed leads to superior optimized controller tuning. In turn, the ALO-MRSO method here can combine a fast transient response with a high steady-state accuracy and robust current regulation. Overall, the ALO-MRSO strategy shows the most balanced and reliable dynamic behavior and is thus especially suitable for real-world PMSM applications that call for precision and stability in different load and speed variations.

Figs. 6, 7, and 8 show the optimized performance of the system during the acceleration and deceleration phases under all four-quadrant operating conditions: positive speed/positive torque, positive speed/negative torque, negative speed/negative torque, and negative speed/positive torque. Each of these figures corresponds to an optimization approach—the ALO, MRSO, and ALO-MRSO, respectively.

Fig. 6, represents the ALO optimized system. From this figure, it can be seen that there is very good responsiveness of speed tracking from the instant of acceleration (0 to around 1.5 sec) and deceleration from 4 to around

4.5 seconds. The actual speed closely follows the reference. However, during the transient phase—around 1.5 and 4.5 seconds—noticeable oscillations and higher ripple are recorded in the torque and current response. Though the system effectively negotiates the four quadrants, for better dynamic stability, smoothness in torque and current can be improved. In contrast, Fig. 7, representative of the system tuned with MRSO, depicts a much smoother dynamic response. The speed tracking performance is as good as the achievements of the ALO-optimized system. Very importantly, the torque and current waveforms reduce oscillations and ripples during both acceleration and deceleration phases. It indicates that the MRSO contributes to better damping characteristics and robustness of current regulation, resulting in a more stable and efficient operation in all four quadrants. It follows from improvements in smoothness of torque and current that MRSO optimizes PI controller gains more effectively than the ALO approach, especially with respect to transient disturbance attenuation.

Fig. 8, which illustrates the hybrid ALO-MRSO approach, demonstrates the superior dynamic performance. The achieved speed tracking is exact, exhibiting minimal deviation from the reference throughout all acceleration and deceleration processes. The most substantial enhancement is observed in the torque and current responses, which are remarkably smooth with almost negligible ripple and spikes across the entire operating

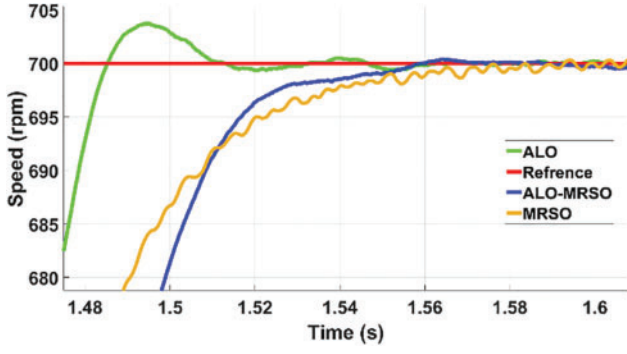


Fig. 9. A comparative analysis of the three optimization techniques.

TABLE II: PERFORMANCE COMPARISON OF OPTIMIZATION METHODS

Method	Rise time (s)	Overshoot (%)	Current ripple (A)
ALO	0.75	0.08	High
MRSO	0.7	0.05	Moderate
ALO_MRSO	0.68	0.03	Low

range, including the challenging four-quadrant transitions. This performance further confirms that the synergistic combination of ALO's global exploration capabilities with MRSO's refined local exploitation strength results in an optimally tuned controller. Thus, the hybrid method effectively delivers a fast transient response coupled with high steady-state accuracy and robust current regulation, making it exceptionally suitable for demanding PMSM applications that require accurate precision and stability.

Fig. 9 highlights the unique strengths and weaknesses in managing the PMSM system performance of the three optimization methods: ALO, MRSO, and the hybrid ALO-MRSO. The ALO approach provides a reasonable dynamic response with acceptable rise times and overshoot and shows moderate torque oscillations and higher current ripples, indicating a limited ability to dampen sudden speed changes. Conversely, MRSO improves upon this by achieving better torque stability and reduced overshoot, owing to its superior local exploitation capabilities. However, the best results are consistently achieved by the hybrid ALO-MRSO approach, which effectively combines the global search efficiency inherent in ALO with the refined convergence characteristics of MRSO. This combination leads to faster response times, minimal overshoot, negligible steady-state errors, and lower torque and current fluctuations. The superior performance of the hybrid strategy underscores its robustness, making it suitable for high-precision, dynamic motor control applications. Table II presents the performance advantages of each method across key control parameters.

7. CONCLUSION

This paper focuses on the control performance of the PMSM with a novel hybrid ALO-MRSO approach based on the field-oriented control framework. The objective of this paper is the effectiveness of the proposed multi-objective cost function, including transient response, steady-state error, and stability constraints, especially when synergistically combined with the ALO-MRSO method

to guide the optimization process. Simulation results demonstrated that though ALO and MRSO, as individual concepts, ensured significant improvements in both speed tracking and torque regulation, the hybrid approach of ALO-MRSO consistently outperformed it for all the cases under consideration. In particular, the hybrid strategy showed faster rise times, lower overshoots, negligible steady-state errors, and greater robustness to parameter variations. The system also ensured a very good dynamic response for different input cases, which, in turn, established the efficacy of the hybrid strategy in applications demanding high-precision motor control.

Collectively, these contributions represent a generalized and robust approach that enables the application of intelligent, heuristic-based optimization to high-performance PMSM controller design. These contributions are summarized in what follows:

- **Development of multi-objective cost function:** This paper introduces the ALO-MRSO-based multi-objective cost function, which integrates transient, steady-state, and stability performance metrics, enabling more robust controller tuning for PMSM speed regulation.
- **Comparative evaluation of optimized techniques:** This study conducts a comparative evaluation of ALO, MRSO, and a novel hybrid ALO-MRSO optimization technique within the context of PMSM control. The study demonstrates that the hybrid approach delivers superior convergence speed, minimal overshoot, and enhanced stability compared to the individual optimizers.
- **Demonstration of enhanced robustness:** this study validates the adaptability of the proposed controller under real-world operating uncertainties by demonstrating enhanced robustness to parameter variations.
- **Simulation-based verification:** The findings of the proposed approach were confirmed through extensive simulations using MATLAB/Simulink software. This provides a clear and comprehensive performance standard for the different operating conditions.

The ALO-MRSO hybrid heuristic optimization offers a strong and effective control solution for PMSM drives. This is particularly important in environments characterized by uncertain parameters and high demand. The flexibility of the proposed control framework makes it an excellent choice for any PMSM-driven systems that require optimal performance, adaptability, and efficiency.

CONFLICT OF INTEREST

The author declares that he does not have any conflict of interest.

APPENDIX

LIST OF ABBREVIATIONS AND SYMBOLS

v_d	d- axis stator voltage
v_q	q-axis stator voltage

R	Stator resistance
i_d	d-axis stator reference current
id^*	q-axis stator reference current
i_q	q-axis stator current
iq^*	q-axis stator current
L_d	d-axis inductance
L_q	q-axis inductance
ω_r	Rotor angular speed
ωm^*	Rotor reference speed
ψ_m	Linkage flux
T_e	Electromagnetic torque
P	Number of poles
T_L	Load torque
Te^*	Average torque
B	Viscous friction coefficient
J	Rotor's moment of inertia
θ_r	Rotor electrical position
$k_1 - k_7$	Weighting factors prioritizing d-q axis current or speed regulation

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